

Host a Website on Amazon S3

AWS Cloud Portfolio Project

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Amazon S3

Static Website Hosting

Bucket Policies

ACLs

Cloud Storage

Introducing Today's Project!

Project overview

In this project, I will demonstrate how to use S3 to host a static website. I'm doing this project to learn how AWS and cloud services and how they can be used to store objects in the cloud and even host websites.

Tools and concepts

Services I used were Amazon S3. Key concepts I learnt include bucket policies, uploading static website files, index.html, bucket endpoint URLs and ACLs and how they control access to a bucket's objects.

Time, challenges, and wins

This project took me approximately 45mins. The most challenging part was resolving the 403 forbidden error. It was most rewarding to the webpage load live and go public.

How I Set Up an S3 Bucket

What I did in this step

In this step we I will open up S3 and then create a storage space inside to start storing website files

How long it took to create the bucket

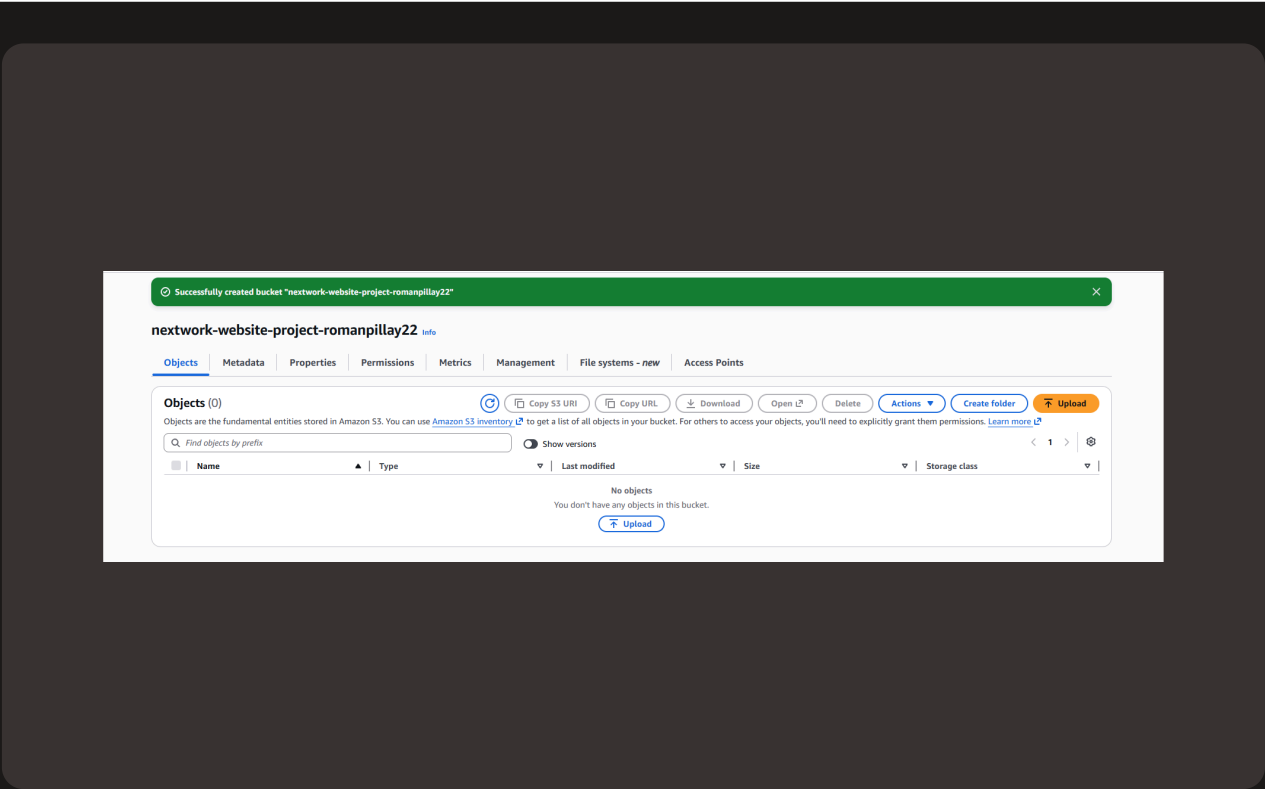
Creating an S3 bucket took me less than 5 minutes i needed to learn a few new concepts like block public access and ACLs, but once that learning is done i can create a bucket in even shorter time in the future.

Region selection

The Region I picked for my S3 bucket was Cape town because its the region thats closest to me. It's best practice to pick the region closest to you because it lowers latency and costs.

Understanding bucket name uniqueness

S3 bucket names are globally unique which means no two buckets can have the same name. They have to be completely unique regardless or region or account.



Upload Website Files to S3

What I did in this step

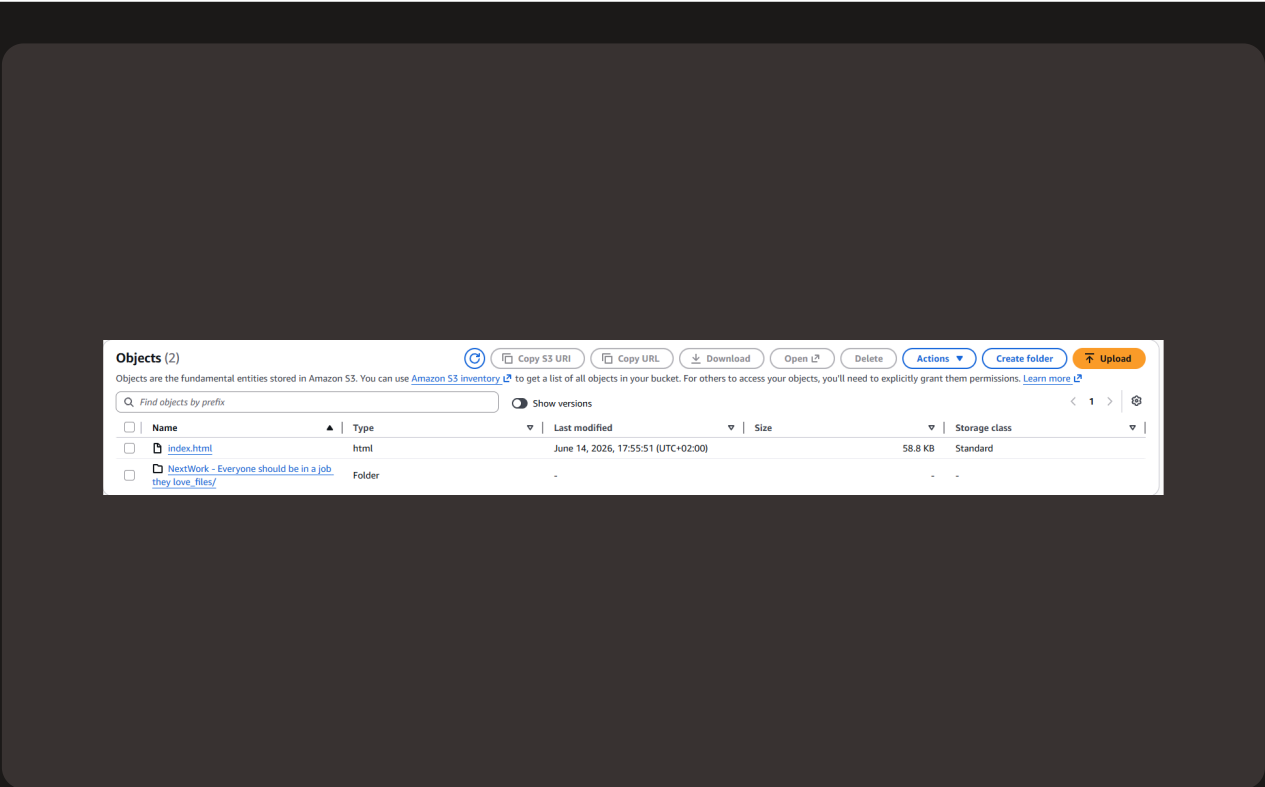
In this step, I will upload website files into my bucket. This is important because no files uploaded into the bucket means no website.

Files I uploaded

I uploaded two files to my S3 bucket - they were an index.html file(this determines the structure of the website) and a folder of images and assets(this will fill in the website).

How the files work together

Both files are necessary for this project as index.html determines the structure but structure alone does not illustrate contents of the website. i.e if index.html says "insert image here" it might not have the actual image to display. I need to supply that image separately, thats why i have multiple files uploaded. The index.html file to direct what is inside the website page, but also a bunch of assets that the website is trying to display.



Static Website Hosting on S3

What I did in this step

In this step, I will make the website available to everyone. This is called static website hosting, and its important because our website files will stay as just files and not turn into website if we don't do this step.

Understanding website hosting

Website hosting means putting our website files on a web server, which is a special computer designed to turn files into a website page that people can visit.

How I enabled website hosting

To enable website hosting with my S3 bucket, I went into the properties tab of our bucket, enabled static website hosting and i also labelled "index.html" as my index document i.e this is the document that we're trying to host.

Access Control Lists (ACLs)

An ACL is a way to configure permission settings inside a bucket. I enabled ACLs so that i can control access to my website files later. There was a pop up mentioning AWS recommends disabling ACLs, but i will keep it enabled to learn how ACLs work and compare it with bucket policies.

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

☐ Disable

☒ Enable

Hosting type

☒ Host a static website

Use the bucket endpoint as the web address. [Learn more](#)

☐ Redirect requests for an object

Redirect requests to another bucket or domain. [Learn more](#)

ⓘ For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document

Specify the home or default page of the website.

index.html

Bucket Endpoints

Understanding bucket endpoint URLs

Once static website is enabled, S3 produces a bucket endpoint URL, which is a URL that takes you to the website you're hosting

What I saw when I tested the endpoint

When I first visited the bucket endpoint URL, I saw a 403 forbidden error. The reason for this error was that objects in a bucket are public by default, even though i have switched off "block all public access" the website files themselves are still completely private. I need to manage their access settings separately. They need to be public files too for the public to see the contents of the website.

403 Forbidden

- Code: AccessDenied
- Message: Access Denied
- RequestId: EPHXT2EW3GBHQPJZ
- HostId: k5BAsXgsCjmwg6tM/lmI3io6NQRyYnbxX51BkJRurWVQNMppAMGxF0OfikPflEOTU1y5uzBuKlc=

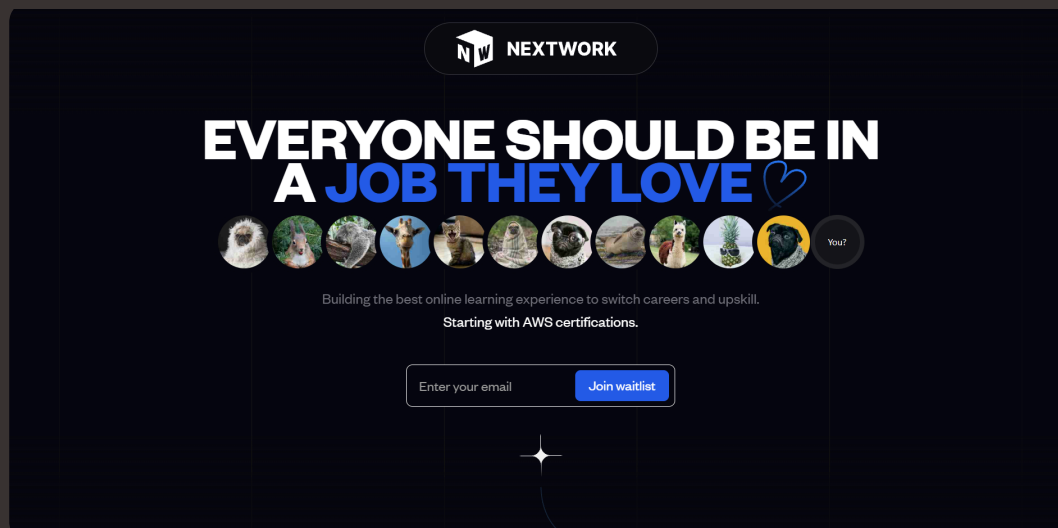
Success!

What I did in this step

In this step, I will go into the files that i uploaded into my bucket and make them public too because this will enable us to view the contents of the website. Once that is done the website is official live.

How I resolved the 403 error

To resolve this 403 Forbidden error, I updated the access settings of the files inside the bucket. Using ACLs i made my bucket files public. Once i checked my S3 bucket endpoint i can see a webpage loaded.



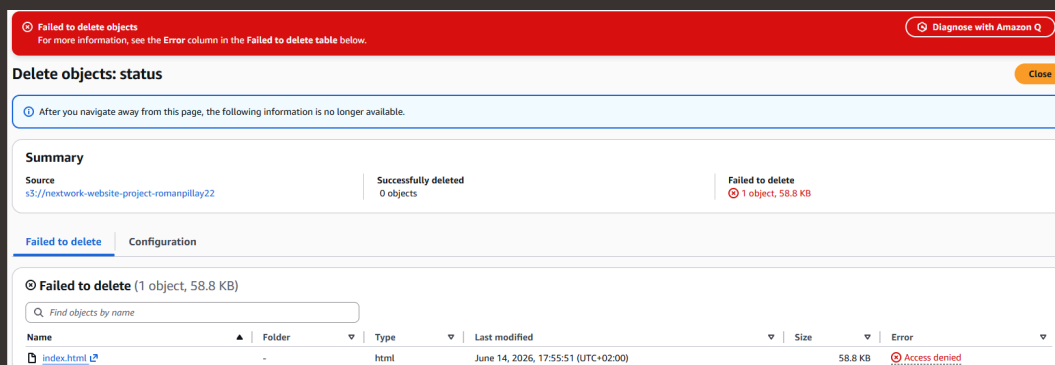
Bucket Policies

What I did in this extension

In this project extension I'm about to use bucket policies to control access to my bucket files. I'm doing this so that i can stop people from deleting objects inside the file.

Understanding bucket policies

An alternative to ACLs are bucket policies, which are rules that determine who is allowed or not allowed to do something. The benefit of using bucket policies is that i can have even greater control of the access of the actions that people are or arent allowed to do. While ACLs are useful for public access to individual access inside the bucket.



What my bucket policy does

My bucket policy denies everyone from deleting the index.html file in the bucket. I tested this by trying to delete index.html and saw a permission denied error. This means the bucket policy worked. It successfully stopped me from deleting objects. i wanted to protect it.